

$$\frac{\text{no. of Correct Pred.}}{\text{Total Pred.}}$$

$$\text{accuracy score} = 0.7 \approx 70\%$$

(Classification)

binary

(Spam / Not Spam)

multi: class
(dog / cat / owl)

Problem:-
→ It does not give the type of errors.

Ex.

① ML → Chest Cancer 99%

② ML → Spam / Not Spam 78% ✓

F → 30%

Not Spam → Spam
Spam → Not Spam

CC → No ✗
No → CC ✓

Confusion Matrix

23 November 2025 16:38

Binary classification

		Predictions	
		1	0
Actual	1	TP	FN
	0	FP	TN

FP → Type 1

FN → Type 2

Actual	Prediction	
	0	1
T	←	→
F	←	→

0 (N)
1 (P)

$$\text{acc. score} = \frac{TP + TN}{TP + FN + FP + TN}$$

$$\text{Setup} = \{0, 1, 2\}$$

Predicted (model)

	0	1	2
Actual (Dataset)			
0	7	0	5
1	2	21	6
2	9	1	13

Accuracy Score:—
$$\frac{\text{Sum (Diagonal elements)}}{\text{Total}}$$

Act. \ Pred	A	B	C
A	a_{11}	a_{12}	a_{13}
B	a_{21}	a_{22}	a_{23}
C	a_{31}	a_{32}	a_{33}

✓ $\frac{1606 \times 100}{10}$

For class A:-

TP:- a_{11}

TN:- $a_{22} \quad a_{23} \quad a_{32} \quad a_{33}$

FP:- $a_{21} \quad a_{31}$

FN:- $a_{12} \quad a_{13}$

For class B

TP:- a_{22}

TN:- $a_{11} \quad a_{13} \quad a_{31} \quad a_{33}$

FP:- $a_{12} \quad a_{32}$

FN:- $a_{21} \quad a_{23}$

$$\text{Precision} :- \frac{TP}{TP+FP} \leftarrow \text{Predicted +ve}$$

(A) Act :-

	Spam	Not Spam
P _u / Spam	100	170
Not Spam	30	700

$$P(A) = \frac{100}{100+170} = \frac{100}{270}$$

$$= \frac{10}{27}$$

$$= 0.37$$

(B) X

	Spam	Not Spam
Spam	100	100
Not Spam	10	700

$$P(B) = \frac{100}{290} = \frac{10}{29}$$

$$= 0.34$$

Type 1

$$P(A) > P(B)$$

$$\text{Recall} = \frac{TP}{TP + FN} \leftarrow \text{total actual +ve}$$

Actual
Detected

1000

800

Not Detected

200

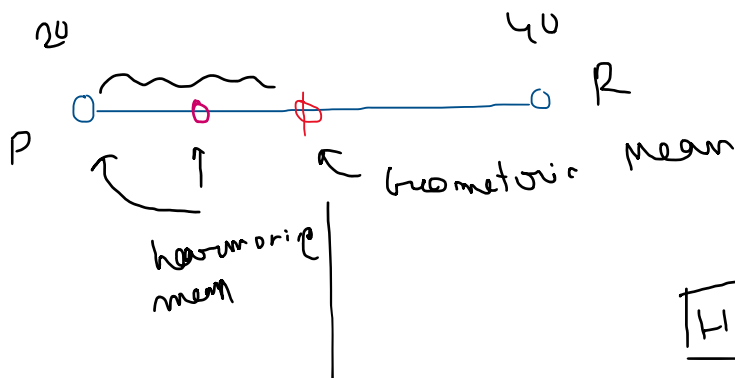
800

P
Yes Cancer
No Cancer

$$\text{Recall} = \frac{1000}{1000 + 800} = \frac{1000}{1800} = \frac{10}{18} = \frac{5}{9} = 0.56$$

Loss Type 2 error

$$F1 = \frac{2PR}{P+R} \quad (\text{harmonic mean})$$



$$GM = 30$$

$$HM = \frac{2 \cdot 20 \cdot 40}{20 + 40} = \frac{1600}{60} = 26.67$$

$$HM < GM$$

$$= \frac{80}{3} = 26.67$$

Multi class Setup:-

	Prediction (Model)			Total
	Dog	Cat	Dwl	
Actual (Data)	Dog	25 11	5 12	40 13
	Cat	0 21	36 21	36 23
	Dwl	4 31	10 21	20 23
	29	45	34	108

Four class Dog:-

$$P_{\text{Dog}} = \frac{TP}{TP + FP} = \frac{25}{25 + 0 + 4} = \frac{25}{29}$$

$$R_{\text{Dog}} = \frac{TP}{TP + FN} = \frac{25}{25 + 5 + 10} = \frac{25}{40} = 0.625$$

$$F_{\text{Dog}} = \frac{2 \cdot \frac{25}{29} \cdot \frac{25}{40}}{\frac{25}{29} + \frac{25}{40}} = \frac{2 \cdot 25 \cdot 25}{25 \cdot 40 + 25 \cdot 29} = \frac{1250}{1250 + 725} = \frac{1250}{1975} = \frac{50}{69}$$

$$F_{dog} = \frac{2 \cdot \frac{40}{25} \cdot \frac{60}{40}}{\frac{25}{25} + \frac{25}{40}} = \frac{2 \cdot \frac{40}{25} \cdot \frac{60}{40}}{25 \times 40} \times \frac{25 \times 40}{25 \times 40}$$

$$\rightarrow 25 \left(\frac{1}{25} + \frac{1}{40} \right)$$

$$= 25 \frac{40+25}{25 \times 40} = 25 \times \frac{65}{25 \times 40}$$

For class Cat

$$P_{cat} = \frac{TP}{TP+FP} = \frac{30}{30+5+10} = \frac{30}{45}$$

$$R_{cat} = \frac{TP}{TP+FN} = \frac{30}{30+0+4} = \frac{30}{34} = 0.88$$

$$F_{cat} = \frac{2 \cdot \frac{30}{45} \cdot \frac{30}{34}}{\frac{30}{45} + \frac{30}{34}}$$

$$= \frac{2 \cdot \frac{30}{45} \cdot \frac{30}{34}}{30 \cdot \frac{34+45}{45 \times 34}} = \frac{2 \cdot \frac{30}{45} \cdot \frac{30}{34} \times \frac{1}{30} \times \frac{45 \times 34}{79}}{\frac{60}{79}}$$

for class Owl

$$P_{owl} = \frac{TP}{TP+FP} = \frac{20}{20+10+4} = \frac{20}{34}$$

$$R_{owl} = \frac{TP}{TP+FN} = \frac{20}{20+10+4} = \frac{20}{34}$$

$$F_{owl} = \frac{2 \cdot \frac{20}{34} \cdot \frac{20}{34}}{\frac{20}{34} + \frac{20}{34}} = \frac{20}{34} = 0.58$$

Recall



Weighted

$$\rightarrow macro = (0.63 + 0.88 + 0.58) \times \frac{1}{3}$$

Weighted macro = (...)

$$0.63 \times \frac{40}{108} + 0.87 \times \frac{34}{108} + 0.59 \times \frac{34}{108}$$

$$\frac{2.11}{3} = \underline{0.703}$$

#		class i
$\sum_{i=1}$	R_i	total